Diffusion Models: From Adding Noise to Generating Images
Understanding Denoising-Based Generation through DDPM and LDM

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Abstract

How can learning to remove noise enable a model to generate new images? This report explains how Gaussian forward corruption, a learned reverse model, and noise-prediction training define an iterative generator. We review two complementary papers: Denoising Diffusion Probabilistic Models (DDPM) establishes a practical pixel-space formulation, while Latent Diffusion Models (LDM) moves denoising into a learned compressed representation. We introduce the original 2015 diffusion framework, then distinguish DDPM's noise-prediction parameterization from direct reverse-parameter estimation. Starting from the variational objective, we explain DDPM's simplified loss, distinguish training from sampling, and illustrate forward noising with a complete scalar calculation and an original five-level image sequence. Published DDPM results demonstrate a sample-quality benefit from objective reweighting; a separately attributed LDM inpainting comparison illustrates the computational benefit of latent diffusion. The discussion emphasizes VAE encoder/decoder roles, briefly introduces conditioning, and separates compression savings from sampler acceleration. Mathematical derivations are provided in the appendices.

1. Introduction

Adding noise destroys image structure, yet learning to remove it can support generation. Across many images and noise levels, a denoiser learns shared statistical structure. A generative process uses this information to produce samples rather than recover one particular corrupted photograph.

Our guiding question is: *How can learning to remove noise enable a model to generate new images?* Our learning objective is to *explain forward noising and reverse denoising, including why latent diffusion reduces computational demands*. These goals organize the report around two complementary papers. DDPM [6] develops the earlier diffusion framework [12] through effective noise-prediction training. LDM [10] asks how to retain that process while reducing repeated computation at image resolution.

We first explain the 2015 probabilistic construction: forward corruption, a learned reverse model, sampling, and variational training. We then isolate DDPM's noise-prediction parameterization and simplified objective. A brief score-matching connection follows the method rather than serving as its prerequisite. We then examine LDM's encoder, decoder, and latent representation. Results distinguish our own forward-noising illustrations from published model measurements. We do not train a denoiser or run a pretrained generator, GPU benchmark, or prompt-adherence experiment.

2. Background and Paper Review

2.1. Generative versus discriminative models

The fundamental distinction between discriminative and generative models lies in what they aim to learn from data. A discriminative model learns to predict a target y given an observation x, typically by estimating $p_\omega(y \mid x)$ or learning a decision boundary between classes. For instance, a cat classifier only needs to identify features that distinguish cats from other objects, without explicitly modeling the distribution of natural images. In contrast, a generative model aims to capture the underlying data distribution, such as $p_\theta(x)$ or the conditional distribution $p_\theta(x \mid y)$. Rather than merely distinguishing categories, it must model the statistical structure and variability of the data, including factors such as shape, texture, lighting, and spatial relationships. This enables the model to generate new samples that resemble the training data, potentially conditioned on labels or text descriptions. Consequently, generative modeling generally requires capturing richer information about the data than is necessary for classification.

2.2. Qualitative Comparison of Generative Models

GANs, autoregressive models, and diffusion models differ in how they learn and sample the data distribution. A *GAN* trains a generator to transform random noise into images

while a discriminator learns to distinguish generated images from real ones [3]. Sampling typically takes one generator pass, but training requires balancing the two competing networks. An *autoregressive model* generates an image one pixel or token at a time, predicting each from the previously generated values [16]. This gives an explicit factorization of the image distribution, although sequential generation can be costly. A *diffusion model* learns to reverse gradual noise corruption and generates an image by repeatedly denoising an initially random state [6]. It refines the whole state across multiple steps, requiring repeated network evaluations during sampling. Figure 5 in Appendix E compares the generative mechanisms of these three families.

2.3. Historical context

The foundations of diffusion-based generative modeling were established by Sohl-Dickstein et al. [12], who introduced a probabilistic framework inspired by nonequilibrium thermodynamics. Their approach consists of a forward diffusion process that gradually destroys the structure of the data by adding noise, followed by a learned reverse process that reconstructs samples from noise. By decomposing a complex generative transformation into a sequence of small, tractable transitions, this framework enables likelihood-based training and probabilistic inference. However, generating samples requires many sequential reverse steps, making inference computationally expensive.

An alternative perspective was developed by Song and Ermon [14] through score-based generative modeling. Instead of directly modeling the data probability density, their approach estimates its score function, defined as the gradient of the log-density. To address difficulties in estimating scores when data lie near low-dimensional manifolds, they introduced multiple levels of Gaussian noise and trained a noise-conditional score network. Samples are then generated using annealed Langevin dynamics, which progressively moves samples toward high-density regions as the noise level decreases.

Ho et al. [6] subsequently introduced Denoising Diffusion Probabilistic Models (DDPM), establishing a connection between diffusion probabilistic modeling and denoising score matching. Their formulation parameterizes the reverse process through a neural network that predicts the Gaussian noise added at each diffusion timestep.

Despite their strong generative performance, conventional diffusion models operate directly in high-dimensional pixel space, resulting in substantial training and sampling costs. To address this limitation, Rombach et al. [10] proposed Latent Diffusion Models (LDM). This architecture incorporates cross-attention mechanisms to support flexible conditioning, including text-guided image generation.

3. Method: From the 2015 Framework to DDPM

The first three subsections explain the Gaussian diffusion framework of Sohl-Dickstein et al. (2015) [12]. We then isolate DDPM's choices of reverse parameterization and training objective [6]. We use x_t and the later $\alpha_t, \bar{\alpha}_t$ notation throughout so that the two formulations can be compared directly.

3.1. 2015: forward process, Gaussian noise, and schedule

A clean image $x_0 \in \mathbb{R}^D$ is gradually corrupted by a Gaussian Markov chain. The Gaussian branch of the 2015 model uses [12, Sec. 2.1, Table C.1]

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t \mathbf{I}\Big), \tag{1}$$

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon^{(t)},$$

$$\varepsilon^{(t)} \sim \mathcal{N}(0, \mathbf{I}), \qquad \alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{j=1}^{t} \alpha_j.$$

Here D counts normalized image coordinates, $\mathbf{I}$ is the identity matrix, and $t \in \{1, \dots, T\}$ is a step index, not physical time. The innovations $\varepsilon^{(t)}$ are independent across steps and coordinates. The dimensionless variance $\beta_t \in (0, 1)$ simultaneously attenuates the previous signal by $\sqrt{1-\beta_t}$ and sets the new noise standard deviation $\sqrt{\beta_t}$. The schedule $(\beta_1, \dots, \beta_T)$ controls how rapidly structure is lost.

The standard Gaussian is stationary under this kernel: if a state has covariance $\mathbf{I}$ and is standard normal, the next state's covariance is $(1-\beta_t)\mathbf{I} + \beta_t \mathbf{I} = \mathbf{I}$. Repeated small steps can therefore transform a complex image distribution toward a simple Gaussian prior. In the 2015 Gaussian experiments, β_1 is fixed to a small value while $\beta_2, \dots, \beta_T$ are learned through the variational bound; the paper also notes that a fixed schedule can work comparably [12, Sec. 2.4.1]. The Gaussian kernel's form is known even when its rates are optimized.

Repeated Gaussian composition gives, with $\bar{\alpha}_0 = 1$,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathbf{I}). \tag{2}$$

This is the same Gaussian marginal later used by DDPM, written in its notation; it is not a new forward-process invention. The aggregate noise ε differs from an individual innovation $\varepsilon^{(t)}$. For small $\bar{\alpha}_T$, the terminal distribution approaches standard normal; a finite schedule need not make that equality exact.

3.2. 2015: learned reverse process and sampling

Let $p_{\text{data}}(x_0)$ denote the unknown data distribution. The original diffusion model learns a finite reverse Markov chain,

starting from a simple Gaussian prior $p(x_T) = \mathcal{N}(0, \mathbf{I})$ [12, Sec. 2.2]:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t),$$
$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)). \quad (3)$$

Although the true reverse conditional depends on the unknown data distribution and is not necessarily Gaussian, sufficiently small diffusion steps motivate approximating each reverse transition with a Gaussian distribution.

For learned intermediate transitions, neural functions f_μ and f_Σ parameterize the mean $\mu_\theta(x_t, t)$ and covariance $\Sigma_\theta(x_t, t)$, respectively. The mean specifies the expected previous state, while the covariance captures uncertainty in the reverse transition. Unlike DDPM's later noise-prediction parameterization, the original model directly estimates these reverse-distribution parameters. The DDPM formulation is discussed in Section 3.4.

To generate a sample, the model draws $x_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively applies the reverse transitions:

$$x_{t-1} = \mu_\theta(x_t, t) + \Sigma_\theta(x_t, t)^{1/2} u_t, \quad u_t \sim \mathcal{N}(0, \mathbf{I}). \quad (4)$$

For diagonal covariance, the matrix square root corresponds to elementwise standard deviations. The original formulation additionally specifies a boundary convention for the final transition at $t = 1$ [12, App. B.2], detailed in Appendix C.5. Through these successive stochastic transitions, the model transforms Gaussian noise into samples approximating p_{data}, without requiring a clean input image.

Figure 1 shows the architecture of both flows: a Gaussian kernel corrupts each forward state, while a neural model parameterizes the reverse distribution from which the previous state is sampled. Appendix F gives step-by-step pseudocode for forward noising and the 2015 Gaussian reverse sampler (Algorithms 1 and 2).

3.3. 2015: Variational Objective and Its Loss Terms

The learning objective is to maximize the probability assigned to real images. Since the exact likelihood integrates over all intermediate states, Sohl-Dickstein et al. optimize a variational lower bound [12, Sec. 2.4]:

$$\mathbb{E}_{p_{\text{data}}}[\log p_\theta(x_0)] \geq K_{2015} = -\sum_{t=2}^{T} \mathcal{D}_t + C_\beta. \quad (5)$$

Equivalently, the loss to minimize is the *negative* bound:

$$\mathcal{L}_{2015} = -K_{2015} = \sum_{t=2}^{T} \mathcal{D}_t - C_\beta,$$
$$\mathcal{D}_t = \mathbb{E}_q[\mathrm{D}_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t))]. \quad (6)$$

Here $\mathbb{E}_q$ averages over training images and the relevant forward states. The target posterior is tractable because x_0 is known:

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}), \qquad t > 1. \quad (7)$$

Each KL penalizes errors in both the reverse mean and covariance. C_β collects entropy terms under the original paper's normalization and endpoint convention. It is constant in the reverse-network weights for a fixed schedule, but must be retained when learning β_t. Appendix C.5 gives these terms and the Gaussian KL.

Three components of the variational loss. To connect this framework to DDPM, write the negative-log-likelihood bound using DDPM's endpoint decoder and notation [6, Eq. 5]:

$$L_{\text{VLB}} = \underbrace{\mathbb{E}_q[\mathrm{D}_{\mathrm{KL}}(q(x_T \mid x_0) \| p(x_T))]}_{L_T} + \sum_{t=2}^{T} \underbrace{\mathbb{E}_q[\mathrm{D}_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t))]}_{L_{t-1}} + \underbrace{\mathbb{E}_q[-\log p_\theta(x_0 \mid x_1)]}_{L_0}. \quad (8)$$

The expectation is distributed over the terms for readability; $\mathbb{E}[-\log p_\theta(x_0)] \leq L_{\text{VLB}}$. L_T measures mismatch between terminal forward noise and the prior. L_{t-1} matches each learned reverse transition to its posterior target. L_0 is the final image reconstruction/decoder likelihood loss. The 2015 formulation uses a prescribed stationary-kernel endpoint; its boundary contributions are incorporated in C_β. Thus Eq. (8) is DDPM's presentation of the inherited variational framework, rather than the original paper's literal formula. We next transform the intermediate L_{t-1} terms.

3.4. DDPM: From Gaussian KL to Noise Prediction

Starting from Eq. (8), Ho et al. [6] make two choices. First, they fix the forward schedule, so L_T is constant during training, and fix the intermediate reverse covariance to $\Sigma_\theta = \sigma_t^2 \mathbf{I}$. The expected Gaussian KL L_{t-1} then reduces to a weighted mean-matching error:

$$L_{t-1} = \mathbb{E}\left[\frac{\|\tilde{\mu}_t - \mu_\theta(x_t, t)\|_2^2}{2\sigma_t^2}\right] + C. \quad (9)$$

Second, instead of predicting $\tilde{\mu}_t$ directly, DDPM exploits the closed-form forward process in Eq. (2), where $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$. Since the added Gaussian noise ε is known during training, the reverse mean can be parameterized using a noise-prediction network:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right). \quad (10)$$

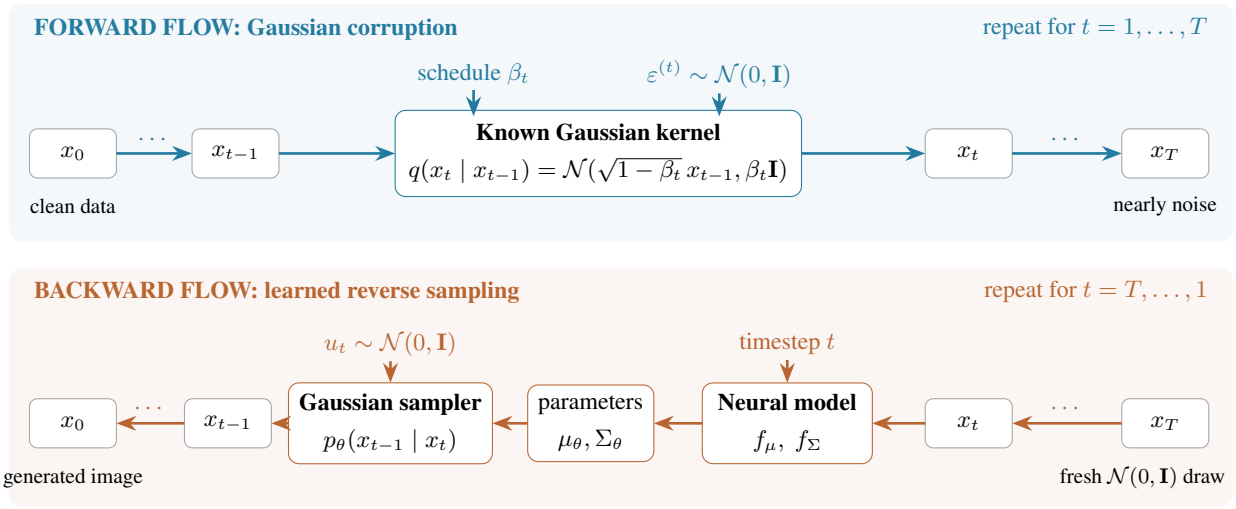


Figure 1. Architecture of the Gaussian diffusion flows [6, 12]. Forward steps (blue) use the noise schedule and Gaussian innovations. Backward steps (orange) start from fresh noise; a neural model predicts reverse-distribution parameters and a Gaussian sampler produces the next state. The displayed neural outputs follow the 2015 formulation; DDPM's noise-based mean parameterization is introduced in Section 3.4. The learned intermediate steps are shown; the final-step boundary is discussed in the text.

Substituting this parameterization into Eq. (9) transforms the mean-matching objective into a weighted noise-prediction loss:

$$L_{t-1} = \mathbb{E}\left[w_t \|\varepsilon - \varepsilon_\theta(x_t, t)\|_2^2\right] + C. \tag{11}$$

Here

$$w_t = \frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1-\bar{\alpha}_t)}, \qquad t > 1. \tag{12}$$

The additive C is independent of θ under these fixed-variance choices. Appendix C.6 gives the substitution in full.

Thus, noise prediction does not replace the underlying variational framework. Instead, it provides an alternative parameterization of the reverse mean, connecting diffusion training to denoising score matching.

3.5. DDPM: Simplified Noise-Prediction Objective

Equation (11) is still an intermediate contribution to the variational bound. DDPM then changes the training objective: it drops the weights w_t and uses the following surrogate [6, Eq. 14]:

$$L_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|_2^2\right]. \tag{13}$$

During training, a clean image x_0, a timestep $t \sim \text{Uniform}\{1, \ldots, T\}$, and $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ are used to construct x_t. The network then learns to predict the injected noise from x_t and t.

Removing w_t changes the relative importance of different noise levels. Ho et al. report that this reweighting improves sample quality, particularly by reducing the emphasis on low-noise reconstruction, although optimizing the original variational bound provides better likelihood estimates [6, Secs. 3.4, 4.1–4.2].

The closed-form forward process also enables efficient training: each update evaluates only one randomly selected timestep. This reduces per-update work compared with unrolling the entire chain; dropping w_t changes training weights rather than the number of sampling evaluations.

L_{simple} is therefore not numerically the full L_{VLB}: L_T is fixed, the intermediate terms are reweighted, and the $t = 1$ noise-MSE term approximates the endpoint decoder treatment described by Ho et al. The exact likelihood bound retains L_0; Appendix C.6 explains that boundary detail.

3.6. DDPM sampling and its score interpretation

Choosing the reverse variance. The 2015 sampler in Section 3.2 learns both a mean and covariance at intermediate steps. In contrast, DDPM 2020 fixes $\Sigma_\theta = \sigma_t^2 \mathbf{I}$ for $t > 1$; the variance is a predetermined function of the timestep, rather than a network output. Ho et al. investigate two choices [6, Sec. 3.2]:

$$\begin{aligned} &\sigma_t^2 = \beta_t \quad \text{or} \quad \sigma_t^2 = \tilde{\beta}_t, \\ &\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t, \qquad t > 1. \end{aligned} \tag{14}$$

Here β_t is the forward transition variance, whereas $\tilde{\beta}_t$ is the variance of the tractable posterior $q(x_{t-1} \mid x_t, x_0)$. The paper reports similar results for these two fixed choices. Their use does not require knowing x_0 during generation, since both are computed from the schedule alone.

Sampling with the chosen variance. Draw $x_T \sim \mathcal{N}(0, \mathbf{I})$ and iterate backward. For $t > 1$, evaluate ε_θ, compute μ_θ from Eq. (10), and sample [6, Algorithm 2]

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t u_t, \quad u_t \sim \mathcal{N}(0, \mathbf{I}). \tag{15}$$

Noise prediction determines the reverse *mean*, while σ_t scales a fresh random draw and controls the transition's stochasticity. At $t = 1$, DDPM sets $u_1 = 0$ and outputs $x_0 = \mu_\theta(x_1, 1)$. This noiseless sampling update differs from the 2015 stationary-kernel endpoint; it does not imply a zero-variance likelihood decoder. Appendices C.6 and F explain these endpoint and sampler distinctions.

Score interpretation. There is also a score interpretation of this parameterization [6, Sec. 3.2]. Let $q_t(x_t)$ be the marginal distribution of noisy data at timestep t. Then

$$s_\theta(x_t, t) := -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}} \approx \nabla_{x_t} \log q_t(x_t). \tag{16}$$

At the ideal MSE optimum $\varepsilon_\theta = \mathbb{E}[\varepsilon \mid x_t, t]$, this is exact under the assumptions in Appendix C.4. It estimates the *noisy* marginal score, not the clean-data gradient at every noise level. No explicit score-evaluation step is added to the algorithm. In conditional LDM, the same interpretation applies to noisy latents given the condition. Appendix C.2 provides optional score background; its role is explanatory rather than a prerequisite for the probabilistic construction above.

3.7. A complete one-dimensional calculation

For one step $t = 1$, choose normalized $x_0 = 0.8$, variance $\beta_1 = 0.1$, and Gaussian realization $\varepsilon = -0.5$. This larger variance makes the arithmetic visible; the image schedule below uses smaller steps. Equation (1) gives

$$\begin{aligned} \alpha_1 &= 1 - 0.1 = 0.9, \\ \sqrt{\alpha_1}\, x_0 &= \sqrt{0.9}\,(0.8) = 0.758947, \\ \sqrt{\beta_1}\, \varepsilon &= \sqrt{0.1}\,(-0.5) = -0.158114, \\ x_1 &= 0.758947 - 0.158114 = \boxed{0.600833}. \end{aligned}$$

Thus $q(x_1 \mid x_0 = 0.8) = \mathcal{N}(0.758947, 0.1)$: its mean is 0.758947, variance 0.1, and standard deviation $\sqrt{0.1} = 0.316228$. Across independent draws, $\mathbb{E}[x_1 \mid x_0] = \sqrt{0.9}\, x_0$ since $\mathbb{E}[\varepsilon] = 0$. The realization 0.600833 differs from that expectation; the conditional variance is β_1.

3.8. Five noise levels on an original small image

Our original 28×28 geometric house has foreground $+1$ and background -1, without an external dataset. The script `code/make_figures.py` simulates one Markov trajectory using Eq. (1) with independent noise at every step. It uses NumPy's default generator, seed 420, $T = 1000$, and the linear schedule $\beta_t = 10^{-4} + (t-1)(0.02 - 10^{-4})/999$, following DDPM's schedule choice [6, Sec. 4].

The last column is a schedule-based signal-to-noise coefficient ratio, also a per-coordinate variance ratio for unit-variance data. This illustrates corruption, not learned denoising or generation.

4. LDM: Latent Space and Efficient Generation

4.1. Why move from pixels to a latent representation?

The diffusion framework supplies a way to generate images, but repeatedly processing a high-resolution pixel grid is expensive. LDM separates learning a useful image representation from learning its generative distribution [10, Secs. 3.1–3.2]. Its first stage compresses images; its second stage applies diffusion to the resulting spatial tensors. The central question is how much computation can be removed while preserving information needed for convincing reconstruction.

A latent is a learned feature representation, not a smaller JPEG or a text description. For RGB input $x_0 \in \mathbb{R}^{H \times W \times 3}$, let $\tilde{z} \in \mathbb{R}^{h \times w \times c}$, where $h = H/f$, $w = W/f$, $f \geq 1$ is the spatial downsampling factor, and c is the latent channel count. For example, an LDM configuration with $f = 4, c = 3$ maps $256 \times 256 \times 3$ to $64 \times 64 \times 3$ [10, Table 8]. This reduces the number of spatial positions by 16, but the end-to-end runtime and memory depend on more than the state shape.

4.2. VAE encoder, decoder, and the compression trade-off

We emphasize LDM's KL-regularized, VAE-like autoencoder. Its encoder predicts a distribution over representations of an input image [8, 10]:

$$\begin{aligned} q_\phi(\tilde{z} \mid x_0) &= \mathcal{N}\big(\mu_\phi(x_0), \operatorname{diag}(\sigma_\phi^2(x_0))\big)\,, \\ \tilde{z} &= \mu_\phi(x_0) + \sigma_\phi(x_0) \odot \xi, \quad \xi \sim \mathcal{N}(0, \mathbf{I}), \\ \hat{x}_0 &= D_\psi(\tilde{z}). \end{aligned} \tag{17}$$

ϕ, ψ are encoder and decoder parameters. The encoder outputs latent means μ_ϕ and positive standard deviations σ_ϕ; diag forms a diagonal covariance and $\odot$ denotes element-wise multiplication. The decoder D_ψ reconstructs an image $\hat{x}_0$. Reparameterization places randomness in ξ, enabling gradient-based training. This posterior noise is different from the diffusion noise added in the second stage.

The encoder must retain spatial and perceptual information; the decoder must learn how to turn it back into image detail. Reconstruction quality therefore limits what the complete generator can preserve. Excessive compression may erase fine text, small objects, or textures that later diffusion cannot faithfully restore.



Figure 2. Original forward-noising illustration at five timesteps. The same trajectory is shown throughout. Values are clipped to $[-1, 1]$ only for display; stored arrays and calculations remain unclipped. The equation and schedule follow [6].

Table 1. Coefficients computed by the supplied script. Signal and noise columns refer to Eq. (2), not measured image quality.

t	$\bar{\alpha}_t$	Signal $\sqrt{\bar{\alpha}_t}$	Noise SD $\sqrt{1-\bar{\alpha}_t}$	$\bar{\alpha}_t/(1-\bar{\alpha}_t)$
0	1.000000	1.000000	0.000000	∞
50	0.971016	0.985401	0.170248	33.5015
200	0.659039	0.811812	0.583919	1.9329
500	0.078587	0.280334	0.959902	0.0853
1000	0.000040	0.006353	0.999980	0.000040

Regularization makes the representation suitable for modeling, but the first stage is not merely a textbook VAE with pixel reconstruction loss. LDM uses perceptual and patch-adversarial reconstruction objectives with a weak KL penalty [10, Sec. 3.1]. Perceptual loss compares visual features; adversarial training encourages locally realistic texture. A textbook VAE objective is given in Appendix D.1 to distinguish these choices. The original paper also studies VQ regularization, so LDM is broader than this KL/VAE variant. Neither weak KL regularization nor successful reconstruction makes clean encoded latents exactly standard normal. The diffusion prior must still learn their distribution.

4.3. Diffusion training and sampling in latent space

After the autoencoder is trained, its parameters are fixed while the latent denoiser is trained. Let $z_0 = a\tilde{z}$, with optional fixed scale $a > 0$ to match latent magnitudes to the noise schedule; $a = 1$ means no rescaling. Diffusion acts on $d_z = hwc$ latent coordinates:

$$z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon_z, \quad \varepsilon_z \sim \mathcal{N}(0, \mathbf{I}_{d_z}), \tag{18}$$

$$L_{\text{LDM}} = \mathbb{E}_{x_0,y,t,\xi,\varepsilon_z}\left[\|\varepsilon_z - \varepsilon_\theta(z_t, t, \tau_\eta(y))\|_2^2\right]. \tag{19}$$

The condition y may be a text prompt, encoded by τ_η with parameters η. The time-conditioned denoiser, typically a U-Net, returns a tensor of the same shape as z_t. The conditional objective follows LDM [10, Eq. 3], here making stochastic encoding explicit. Posterior-mean encoding omits ξ.

For generation, start with latent Gaussian noise and repeatedly apply a compatible reverse sampler using the conditional predictor. Decode the final latent once as $D_\psi(z_0/a)$, undoing the training scale. Pure text-to-image generation needs no real input image or image encoder. The encoder is needed to form training representations, whereas the decoder converts a generated representation into pixels. Figure 3 makes these roles explicit.

4.4. Conditioning and the complete pipeline

LDM uses cross-attention to make denoising depend on a condition such as text [10, Sec. 3.3]. Spatial image features supply queries; prompt-token embeddings supply keys and values. Learned attention weights retrieve text information useful for the current features. Thus a prompt changes the denoising predictions throughout the sampling loop; it is not pasted onto the decoded image. The matrix formulation and an optional toy calculation are confined to Appendix D.2.

Classifier-free guidance can strengthen conditional influence by using $\varepsilon_u + s(\varepsilon_c - \varepsilon_u)$, where $\varepsilon_u, \varepsilon_c$ are unconditional and conditional predictions and $s \geq 0$ is the guidance scale [5]. Here $s = 1$ is ordinary conditional sampling. Stronger guidance can trade diversity for conditional fidelity; it does not guarantee correct counting, spatial relations, or attribute binding. Standard CFG uses two denoiser evaluations per step. Details are in Appendix D.3.

The two papers therefore address complementary questions: DDPM learns how iterative denoising can generate an image distribution; LDM learns a representation in which that iteration is less expensive. Compression reduces repeated spatial work. Faster samplers such as DDIM instead

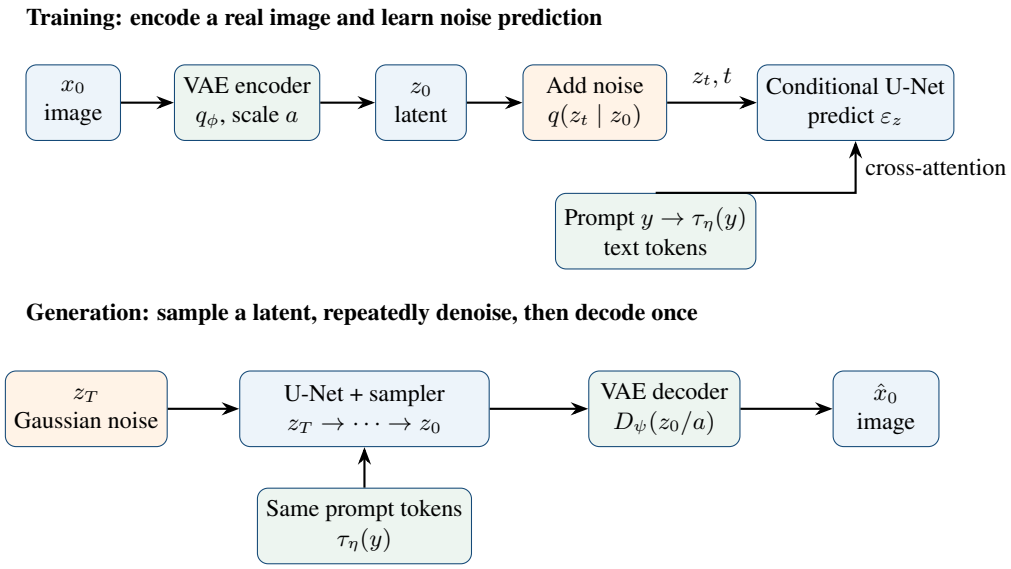


Figure 3. Original latent diffusion pipeline, based on [10, Fig. 3 and Sec. 3]. The VAE is learned first and frozen for diffusion training; its decoder runs after the denoising loop. Pure text-to-image generation does not require a real input image or the image encoder.

reduce the number of evaluations [13]; the two ideas can be combined without treating them as the same speedup.

5. Results and Discussion

5.1. Original illustrations and evidence boundaries

Our own results are the scalar calculation in Section 3.7 and the forward trajectory in Section 3.8. Figure 2 shows the synthetic house losing recognizable structure as the signal multiplier falls from 1 to 0.00635. Table 1 relates the sequence to the schedule. One random trajectory, especially with display clipping, does not establish a population-level image-quality trend.

These calculations illustrate forward corruption, not successful learned reversal. No pretrained generation, denoising benchmark, GPU runtime, peak-memory, or prompt-adherence measurement was performed. The following two comparisons are published results from different studies and tasks; they must not be combined into a DDPM-versus-LDM quality ranking.

5.2. Published DDPM evidence for the objective choice

Table 2 reports Ho et al.'s unconditional CIFAR-10 comparison with epsilon prediction and fixed isotropic reverse variance [6, Tables 1–2]. The simplified objective lowers FID from 13.51 to 3.17, while its reported negative-log-likelihood upper bound is slightly higher. This demonstrates a sample-quality/codelength trade-off in the authors' setting, rather than equivalent optimization of the original likelihood bound.

Table 2. Published DDPM objective comparison [6, Table 1]. NLL upper bounds are in bits/dimension; lower values are better. These are not measurements from our team.

Objective	FID	NLL bound
Variational bound L	13.51	3.70
Simplified L_{simple}	3.17	3.75

FID (Fréchet Inception Distance) compares generated and reference feature-distribution statistics [4]; it does not directly measure every image's quality or prompt adherence. DDPM computes these CIFAR-10 FIDs against training images; its reported test-reference FID for the simplified model is 5.24 [6, Sec. 4.1]. Stating the reference set matters. The ablation supports the combination of epsilon prediction and the simplified objective, not a claim that any unweighted reconstruction loss improves any generative model.

5.3. A published pixel-versus-latent setting

We select the *Places inpainting* comparison in Rombach et al. [10, Sec. 4.5, Table 6]: pixel-space LDM-1 versus LDM-4 with a KL-regularized first stage and attention. Their denoisers have a matched parameter count. The published protocol trains on 256×256 crops; the supplementary protocol specifies fixed validation masks and a 2,000-image validation set. We preserve the table's two sampling resolutions and six-epoch endpoint.

Our arithmetic gives sampling-throughput ratios $0.97/0.26 = 3.73$ at 256^2 and $0.34/0.07 = 4.86$ at 512^2. The epoch-time ratio is $20.66/7.66 = 2.70$ and training-

Table 3. Published inpainting comparison, transcribed from [10, Table 6]. These are the paper's measurements, not a benchmark run by our team. Higher throughput and lower FID are better.

Criterion	Pixel LDM-1	LDM-4 (KL, with attention)
Training throughput (samples/s)	0.11	0.32
Sampling throughput at 256^2 (samples/s)	0.26	0.97
Sampling throughput at 512^2 (samples/s)	0.07	0.34
Training + validation (hours/epoch)	20.66	7.66
Validation FID after epoch 6 (2k samples)	24.74	15.21
Peak memory (GB)	Not reported in this table	

throughput ratio $0.32/0.11 = 2.91$. The reported FID is 9.53 points lower, a relative decrease of 38.5%; this percentage is *not* a percentage increase in perceptual quality.

The result supports an efficiency–quality advantage for this configuration and training endpoint. Six equal epochs do not imply equal wall-clock training budgets or equal convergence. Moreover, the table flags GPU-setting/batch-size differences from the paper's Figure 7; its ratios should not be universalized. Separate autoencoder pretraining is an additional cost not isolated by this table. Inpainting uses spatial conditions, so this experiment does not itself establish a text-prompt-control advantage.

5.4. Speed, quality, memory, and control: explicit criteria

Table 4 connects the benchmark to the mechanisms developed in Section 4. Its memory and control entries are analytical comparisons, not unreported empirical measurements.

Where speed comes from. An analytical latency decomposition is

$$\mathcal{T}_{\text{gen}} \approx \mathcal{T}_{\text{text}} + K\, r\, \mathcal{T}_{\text{U-Net}} + \mathcal{T}_{\text{decode}}, \tag{20}$$

where every $\mathcal{T}$ is measured in seconds, K is the number of sampling updates, and r is the effective number of denoiser evaluations per update (1 without standard CFG, 2 with it, before batching effects). The final term is absent in pixel diffusion. Latent compression mainly reduces the cost of a U-Net evaluation; DDIM can reduce K [13]. Fewer updates, higher guidance, and greater compression are separate design choices with different quality consequences.

Why a smaller state is not a guaranteed memory ratio. At 256^2, RGB has $196{,}608$ coordinates; the illustrative $64 \times 64 \times 3$ latent has $12{,}288$, giving a $16{:}1$ state-element ratio. Generally it is $3f^2/c$. At fixed feature width, spatial activation storage also benefits from fewer positions, but model weights, attention, batching, and the full-resolution decoder affect the peak. Therefore neither a $16\times$ runtime speedup nor a $16\times$ total-memory reduction follows from this arithmetic.

Quality and applicability. Moderate compression can make iterative generation practical at larger resolutions. Tasks needing exact text, tiny structures, or pixel-faithful reconstruction must also inspect autoencoder error. Stronger guidance and lower FID cannot substitute for checking requested object attributes and relations. These distinctions explain why the favorable inpainting result is useful evidence without settling every application.

5.5. Training data, bias, copyright, and limitations

Data dependence. The VAE learns what can be represented, the denoiser learns what is likely, and the text encoder learns which language features connect to imagery. Errors or underrepresentation at any stage can propagate to outputs. Large image–text corpora such as LAION-5B illustrate the scale of this dependency [11]; they are not interchangeable with the specific datasets used in every original LDM experiment. Caption noise, duplicate images, language imbalance, and missing provenance are distinct concerns. Our analytical recommendation is to document dataset versions, sources, filters, and evaluation splits, and audit duplicates and underrepresented concepts.

Bias and prompt control. Bianchi et al. [1] report demographic stereotype amplification in text-to-image systems, including prompts that do not explicitly name demographic groups. Latent compression does not remove these associations. Because guidance amplifies conditional preferences and can reduce diversity, stronger prompt adherence should not be assumed to imply fairer representation. Evaluation should compare distributions across multiple prompts, groups, and seeds; prompt rewriting alone is insufficient evidence of mitigation.

Memorization and copyright. Carlini et al. [2] demonstrate extraction of memorized training examples from diffusion models. This establishes a possible failure mode, not that every output is copied. Dataset availability, permission to use underlying works, memorization risk, and rights in an output are separate questions. The U.S. Copyright Office's 2025 pre-publication analysis distinguishes data collection, training, outputs, and licensing [15]; it is a U.S. policy source, not Vietnamese law. Compression is no technical guarantee against reproduction. Our recommendations are documented provenance, appropriate permissions, deduplication, and similarity checks for sensitive use cases;

Table 4. Pixel versus latent diffusion. The concrete latent setting is LDM-4 with KL regularization. Architecture-based analysis is informed by [5, 6, 10]; measured values are confined to Table 3.

Criterion	Pixel diffusion	Latent diffusion (KL, $f = 4$)
Speed	Repeatedly processes a pixel-resolution state	Smaller repeated states; one final decode. The published inpainting throughput is higher in the selected setting
Quality	No separate autoencoder bottleneck; still limited by denoiser, data, and sampling	Compression can retain useful structure but lose detail or introduce reconstruction artifacts
Memory	Larger spatial states and potentially larger feature maps	Fewer latent positions; autoencoder/text-encoder weights and decoder activations add memory. End-to-end peak memory needs measurement
Control	Can support text conditioning and guidance when trained for it	Cross-attention and guidance support flexible prompts; compression itself does not guarantee better semantic control

none alone certifies legal compliance.

Limits of this report. Our original evidence covers a scalar calculation and one synthetic forward trajectory. The model evidence comprises published DDPM objective results and an LDM inpainting comparison, not our own matched text-to-image benchmark. We have no measured peak-memory values or prompt-adherence scores. A future study should hold hardware, precision, batch size, sampler, step count, guidance, resolution, and dataset fixed, report both autoencoder and denoiser costs, and evaluate quality and control separately.

6. Conclusion

Learning to remove noise enables generation by fitting a sequence of reverse conditional distributions. The 2015 diffusion framework supplies forward Gaussian corruption, learned reverse mean/covariance functions, and variational training against tractable posterior targets. Starting from a new Gaussian draw and composing the reverse transitions then produces an image sample. DDPM retains this construction while fixing variance choices, parameterizing the reverse mean through predicted noise, and simplifying the training objective. Its published ablation supports a sample-quality benefit from that objective choice, distinct from likelihood optimization or fewer sampling steps. The score connection explains the parameterization without adding a separate computational stage.

LDM preserves this denoising principle while operating in a learned latent representation. The encoder/decoder trade reconstruction fidelity for reduced repeated spatial computation. Published inpainting evidence supports an efficiency benefit in a specific configuration, with autoencoder costs and information loss still relevant. Together, DDPM and LDM develop the original framework through better parameterization and more efficient representation; neither guarantees semantic control, fairness, or unrestricted data use.

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A. Requirement-to-Evidence Map

This map concerns the written report. The assignment's separate slide, oral-presentation, balanced-participation, and Q&A requirements require presentation deliverables and cannot be fulfilled by a report alone. The main text occupies pages 1–9; references and appendices are separate.

Table 5. Coverage of the supplied CS420 S06 seminar brief.

Task	Required content	Location / evidence
1	Generative versus discriminative; GAN, autoregressive, diffusion	Sections 2.1, 2.2; Figure 5 (Appendix E)
2	Forward process, beta schedule, Gaussian noise, noise predictor, reverse sampling; both-direction diagram	Section 3; Eqs. (1)–(15); Figure 1
3	Manual 1D forward step, symbols, stochastic expectation	Section 3.7; $x_1 = 0.600833$; conditional mean and variance
4	Small image at five noise levels; optional pretrained demonstration	Section 3.8; Figure 2; Table 1; executable code and arrays. Pretrained generation was not performed
5	VAE encoder/decoder, latent operations, cross-attention, pipeline	Sections 4.2–4.4; Eqs. (17)–(19); Figure 3
6	Speed, quality, prompt control, data, copyright, bias; explicit pixel/latent comparison	Sections 4.4, 5.3–5.5; Tables 3–4

Required representations. Equation (1) is the Gaussian forward transition; Eq. (13) is the simplified noise-prediction objective. The worked example gives a complete substitution, result, and interpretation. Appendix C records ranges, units, and modeling assumptions.

Paper reading and synthesis. DDPM's forward/reverse method and contribution are reviewed in Section 3, distinguishing the 2015 foundation from DDPM's parameterization and loss choices. LDM's representation, method, conditioning, contributions, and differences are developed in Section 4. Sections 5.2 and 5.3 identify the source tables for the DDPM objective and LDM inpainting comparisons. The report includes Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References. English is used according to the commissioning instruction, which overrides the brief's Vietnamese-language request.

B. Reproducibility, Evaluation Protocol, and Attribution

B.1. Reproduce the original illustration

The Overleaf project already contains the generated figure; Python is not needed to compile the report. To regenerate it locally:

```
python -m pip install -r code/requirements.txt
python code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-
    error main.tex
```

The figure uses a synthetic geometric image, not MNIST or another dataset. The script records the seed, schedule coefficients, library versions, scalar example, and benchmark-ratio arithmetic in `figures/analysis.json`. It saves unclipped frames, timesteps, the schedule, and cumulative coefficients in `noise_arrays.npz`, and summary coefficients in `noise_levels.csv`. The delivered run used NumPy 2.3.5 and ReportLab 4.4.9. The sequence is one sampled Markov path; identical seeds and library versions reproduce it. No image-generation checkpoint is involved.

B.2. A future prompt and runtime study (not executed)

Select and record exact checkpoint identifiers, revisions, dataset licenses, hardware, software versions, numerical precision, resolution, and batch size. Keep each fixed within a comparison and record warm-up policy. For latent models, measure text encoding, denoising, decoding, and end-to-end latency separately, and record peak allocated memory over the entire pipeline.

Use paired prompts such as *a red cube to the left of a blue sphere* and *a green cube to the left of a blue sphere*; separately swap the spatial relation. Predefine at least 20 seeds per condition and inspect all samples. Vary sampling steps and guidance in separate sweeps. Report object presence, color binding, relation correctness, and diversity; do not select only attractive examples. This is a proposed evaluation design, not empirical evidence of a model's prompt capability.

B.3. Provenance and AI assistance

Figures 1, 3, and 4 are original TikZ schematics explaining cited methods; no paper figure is reproduced. Figure 2 and Table 1 are generated by the supplied code. Table 2 reproduces selected DDPM Table 1 values. Table 3 is transcribed from LDM Table 6; ratios are our arithmetic. Table 4 is an analytical synthesis with sources in its caption. Figure 5 is an illustration included with the project; related method sources are cited in its caption.

The document uses the requested ICCV 2025 author kit [7], with its official style and bibliography files. It uses 10-point type, two columns, and US Letter paper, following the new template request. The author block retains HCMUS and CS420 information and three name/ID placeholders. Template provenance is included in the project.

OpenAI Codex [9] assisted with drafting, equations, code, diagram construction, source checking, and LaTeX preparation. Preparation checks included consulting the original DDPM/LDM equations and DDPM Tables 1–2 and LDM Table 6, recalculating the numerical examples and ratios, and compiling and visually reviewing the PDF. The named student authors should review and accept the final manuscript and replace the author placeholders before submission. No human study, new trained model, or pretrained generation benchmark is claimed.

C. Notation, Assumptions, and Additional Derivation

C.1. Ranges and units

Images are normalized and dimensionless. Clean pixels in the illustration lie in $[-1, 1]$; noisy states, latent values, and predicted noise are real-valued and need not stay within that interval. There is no physical time unit in the diffusion index.

Table 6. Notation used in the equations; meanings are also introduced beside the corresponding equations in the main text.

Symbols	Meaning, range, and units
$x_0, x_t, \hat{x}_0$	Clean/noisy/reconstructed image; vector in $\mathbb{R}^D$ or an equivalent image tensor; normalized intensity units
D, H, W, h, w, c, d_z	Positive integer dimensions. $D = 3HW$ for RGB, $d_z = hwc$; H, W count pixels, h, w count latent positions
t, T, j	Integer timestep, terminal step, and product index; $1 \le t \le T$, $1 \le j \le t$
$\beta_t, \alpha_t, \bar{\alpha}_t$	Dimensionless schedule coefficients; $0 < \beta_t < 1$, $\alpha_t = 1 - \beta_t$, $0 < \bar{\alpha}_t \le 1$
$q, p_\theta, p_{\text{data}}$	Fixed forward transition, learned reverse/generative distribution, and data distribution, respectively
$\rho, \tilde{x}, q_\rho, s$	Positive Gaussian perturbation SD, perturbed image, noisy marginal, and score in Appendix C.2; s there is a vector field, distinct from the scalar CFG scale
$K_{2015}, \mathcal{D}_t, C_\beta$	Original log-likelihood lower bound, expected posterior KL, and schedule-dependent entropy terms
$\mathcal{L}_{2015}, L_{\text{VLB}}$	Minimized negative 2015 bound and full DDPM variational negative-log-likelihood bound, in nats
L_T, L_{t-1}, L_0	Expected prior KL, intermediate posterior-to-model KL, and final decoder negative log likelihood, in nats
$w_t, \tilde{\beta}_t, \tilde{\mu}_t$	Positive loss weight; forward-posterior variance and mean, for $t > 1$; see Appendix C.6
$\mathcal{N}(\mu, \Sigma), \mathbf{I}$	Gaussian with mean μ and covariance Σ; identity covariance of the relevant dimension
$\varepsilon, \varepsilon_z, u, \xi$	Independent standard-normal draws in their respective sampling operations; each coordinate has mean 0 and variance 1
Σ_θ	Learned positive diagonal reverse covariance in the 2015 image model; fixed to $\sigma_t^2 \mathbf{I}$ in the DDPM setup discussed
$\sigma_t, \mu_\phi, \sigma_\phi$	Reverse noise standard deviation $\sigma_t \ge 0$; VAE posterior mean and positive standard deviations
$\theta, \phi, \psi, \eta, \omega$	Trainable parameter arrays of denoiser, VAE encoder, decoder, condition encoder, and illustrative classifier
$\tilde{z}, z_0, z_t, f, a$	Raw/clean-scaled/noisy latent; spatial factor $f \ge 1$, with H, W divisible by f; fixed dimensionless scale $a > 0$
$y, \tau_\eta(y), F, C$	Condition, encoded condition, U-Net features, and token embedding matrix. Continuous embeddings are real-valued
n, m, d_u, d_y, d_a, d_v	Positive integer counts and widths for cross-attention; see Appendix D.2 for all matrix shapes
Q, K, V, A, O	Query, key, value, attention-weight, and output matrices; $A_{ij} \in [0, 1]$ and $\sum_j A_{ij} = 1$
$s, w, p_{\text{drop}}, \varnothing$	CFG scale $s \ge 0$, alternative scale $w = s - 1$, dropout probability in $(0, 1)$, and null condition
$K, r, \mathcal{T}$	In the cost model only, K counts updates (not attention keys), r counts evaluations/update, and $\mathcal{T}$ is seconds
$L_{\text{simple}}, L_{\text{LDM}}$	Expected summed squared noise-prediction errors; dimensionless squared normalized-coordinate units
$\mathbb{E}, \lVert \cdot \rVert_2, \mathrm{D}_{\text{KL}}$	Expectation, Euclidean norm, and KL divergence; natural logarithms make KL and negative log likelihood use nats

C.2. Optional background: scores and Gaussian perturbation

For a smooth positive density p, define its score as

$$s(x) = \nabla_x \log p(x). \tag{21}$$

This vector gives the direction of steepest local increase in log density, not a scalar image-quality rating. For $p(x) = \mathcal{N}(0,1)$, $\log p(x) = -x^2/2 + C$ and $s(x) = -x$. Score-based sampling combines density-directed drift with randomness. For example, a discretized Langevin update is [14]

$$v^{(k+1)} = v^{(k)} + \frac{\delta}{2} s(v^{(k)}) + \sqrt{\delta}\,\zeta^{(k)}. \tag{22}$$

Here $v^{(k)}$ is the current state, k counts updates, $\delta > 0$ is a step size, and $\zeta^{(k)} \sim \mathcal{N}(0, \mathbf{I})$ are independent draws. The stochastic term supports distributional exploration rather than only mode seeking. Finite-step sampling is approximate; convergence requires suitable step-size limits and regularity conditions. This supplies context for the score-based literature rather than an additional algorithm required by the DDPM construction in the main text.

Gaussian smoothing addresses difficulties in estimating scores when data concentrate near low-dimensional subsets of image space [14]. A simple additive example is

$$\tilde{x} = x_0 + \rho\varepsilon, \quad x_0 \sim p_{\text{data}}, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}), \quad \rho > 0. \tag{23}$$

ρ is a noise standard deviation. Its known Gaussian conditional $q_\rho(\tilde{x} \mid x_0) = \mathcal{N}(x_0, \rho^2\mathbf{I})$ has score $-\varepsilon/\rho$. The marginal $q_\rho(\tilde{x})$ averages over clean data; learning its score requires the conditional-to-marginal identity below. Different noise levels define different smoothed distributions. DDPM instead scales the clean signal as well, as in Eq. (2).

C.3. Why the forward marginal is available in closed form

Assume independent standard-normal innovations $\varepsilon^{(1)}, \varepsilon^{(2)}$ and let x_0 be fixed. Applying Eq. (1) twice gives

$$\begin{aligned} x_1 &= \sqrt{\alpha_1}x_0 + \sqrt{\beta_1}\varepsilon^{(1)}, \\ x_2 &= \sqrt{\alpha_2\alpha_1}x_0 + \sqrt{\alpha_2\beta_1}\varepsilon^{(1)} + \sqrt{\beta_2}\varepsilon^{(2)}. \end{aligned}$$

The two independent noise terms have total variance $\alpha_2\beta_1 + \beta_2 = \alpha_2(1-\alpha_1) + (1-\alpha_2) = 1 - \alpha_2\alpha_1$. A sum of independent Gaussians is Gaussian. Repeating this argument yields Eq. (2), with mean $\sqrt{\bar{\alpha}_t}x_0$ and covariance $(1-\bar{\alpha}_t)\mathbf{I}$ [6].

Gaussian innovations are independent across coordinates and steps, but the noisy images themselves have correlations inherited from x_0. When $\bar{\alpha}_T$ is close to zero, dependence on the starting image is weak and a standard-normal starting prior is a useful approximation. A finite schedule need not make it exactly equal to that prior. The learned Gaussian reverse transition is a modeling approximation motivated by small forward steps; no finite network is assumed to know the exact reverse conditional.

C.4. Denoising score matching and the MSE optimum

Fix a nonzero noise level and write $x_t = a_t x_0 + b_t\varepsilon$ with $a_t, b_t > 0$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ independent of x_0. The additive example of Appendix C.2 uses $a_t = 1, b_t = \rho$; DDPM uses $a_t = \sqrt{\bar{\alpha}_t}$ and $b_t = \sqrt{1-\bar{\alpha}_t}$. These a_t coefficients are unrelated to the fixed latent scale a. Let q_t be the marginal over noisy data. Gaussian smoothing makes it positive and smooth; assume sufficient integrability to exchange integration and differentiation. Then

$$\begin{aligned} q_t(x_t) &= \int p_{\text{data}}(x_0) q(x_t \mid x_0)\, dx_0, \\ \nabla \log q_t(x_t) &= \int q(x_0 \mid x_t, t) \nabla \log q(x_t \mid x_0)\, dx_0. \end{aligned} \tag{24}$$

The conditional Gaussian has score

$$\nabla \log q(x_t \mid x_0) = -\frac{x_t - a_t x_0}{b_t^2} = -\frac{\varepsilon}{b_t}. \tag{25}$$

Taking its posterior expectation in Eq. (24) gives

$$\nabla \log q_t(x_t) = -\frac{\mathbb{E}[\varepsilon \mid x_t, t]}{b_t}. \tag{26}$$

This is the conditional-to-marginal identity underlying denoising score matching [14, 18].

For any vector-valued predictor g and finite second-moment target Y, condition on an input X. With $m(X) = \mathbb{E}[Y \mid X]$, expand

$$\begin{aligned} \mathbb{E}[\|Y - g(X)\|^2 \mid X] = \mathbb{E}[\|Y - m(X)\|^2 \mid X] \\ + \|m(X) - g(X)\|^2. \end{aligned} \tag{27}$$

The cross term vanishes because $\mathbb{E}[Y - m(X) \mid X] = 0$. Thus the ideal MSE predictor is $m(X)$. Setting $Y = \varepsilon$ and $X = (x_t, t)$ justifies the interpretation in Section 3.6. A finite trained network only approximates that optimum.

If $\varepsilon_\theta = -b_t s_\theta$, a conditional-target score loss satisfies

$$\mathbb{E}\|s_\theta + \varepsilon/b_t\|^2 = \frac{1}{b_t^2}\mathbb{E}\|\varepsilon - \varepsilon_\theta\|^2 \tag{28}$$

at a fixed t. Across timesteps, weights must be accounted for: unweighted noise MSE corresponds to weighting this score loss by b_t^2. The two unweighted losses should not be identified without that factor. The identity is used only for $b_t > 0$, not at the clean endpoint.

C.5. The original 2015 bound and learned covariance

For a Gaussian diffusion model, the general trajectory lower bound is

$$\mathbb{E}[\log p_\theta(x_0)] \geq \mathbb{E}_q[\log p_\theta(x_{0:T}) - \log q(x_{1:T} \mid x_0)]. \tag{29}$$

Sohl-Dickstein et al. [12, Sec. 2.4, App. B] rewrite this, under their normalization and stationary-kernel endpoint conventions, as

$$\begin{aligned} K_{2015} &= -\sum_{t=2}^{T} \mathcal{D}_t + C_\beta, \\ \mathcal{D}_t &= \mathbb{E}_q\left[\mathrm{D}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t)\right)\right], \\ C_\beta &= H_q(X_T \mid X_0) - H_q(X_1 \mid X_0) - H_\pi(X_T). \end{aligned} \tag{30}$$

H denotes differential entropy in nats and $\pi = \mathcal{N}(0, \mathbf{I})$. The stationary-prior cross-entropy simplification uses the paper's normalization of the data. For the isotropic Gaussian forward chain, $H_q(X_t \mid X_0) = \frac{D}{2} \log[2\pi e(1 - \bar{\alpha}_t)]$. These terms depend on the schedule; treating C_β as constant is valid with respect to reverse parameters at a fixed schedule, not when optimizing the forward rates.

In particular, the 2015 endpoint is prescribed as

$$p(x_0 \mid x_1) = T_\pi(x_0 \mid x_1; \beta_1) = \mathcal{N}(\sqrt{1 - \beta_1} x_1, \beta_1 \mathbf{I}), \tag{31}$$

where T_π is the stationary Gaussian kernel. This removes the learned endpoint term from its KL sum. It should not be confused with DDPM's endpoint likelihood convention or its final noiseless display update.

For $q = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t \mathbf{I})$ and learned diagonal $p_\theta = \mathcal{N}(\mu_\theta, \mathrm{diag}(v_{\theta,1}, \ldots, v_{\theta,D}))$, with $v_{\theta,i} > 0$, the per-example KL is

$$\begin{aligned} \mathrm{D}_{\mathrm{KL}}(q \| p_\theta) = \frac{1}{2} \sum_{i=1}^{D} \Bigg[&\log \frac{v_{\theta,i}}{\tilde{\beta}_t} - 1 \\ &+ \frac{\tilde{\beta}_t + (\tilde{\mu}_{t,i} - \mu_{\theta,i})^2}{v_{\theta,i}} \Bigg]. \end{aligned} \tag{32}$$

Learning covariance therefore changes more than a mean-MSE target. Only after fixing reverse variance do the variance-dependent pieces become constant in θ, giving the mean-MSE expression used in DDPM's parameterization argument. The explicit Gaussian posterior parameters are given next.

C.6. From the variational bound to weighted noise MSE

DDPM retains the joint generator and Gaussian forward construction with a fixed schedule [6, Secs. 2–3]:

$$\begin{aligned} p_\theta(x_{0:T}) &= p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \\ q(x_{1:T} \mid x_0) &= \prod_{t=1}^{T} q(x_t \mid x_{t-1}). \end{aligned} \tag{33}$$

Jensen's inequality bounds negative log likelihood by $L = \mathbb{E}_q[\log q(x_{1:T} \mid x_0) - \log p_\theta(x_{0:T})]$. Here L is the L_{VLB} introduced in Eq. (8). Rearranging gives

$$\begin{aligned} L = \mathbb{E}\Bigg[&\mathrm{D}_{\mathrm{KL}}(q(x_T \mid x_0) \| p(x_T)) \\ &+ \sum_{t=2}^{T} \mathrm{D}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t)\right) \\ &- \log p_\theta(x_0 \mid x_1) \Bigg]. \end{aligned} \tag{34}$$

The terminal term is fixed when the schedule and prior are fixed. The endpoint term is a reconstruction/decoder likelihood; DDPM's original implementation uses a discretized decoder for image integers. It is not simply another intermediate Gaussian KL.

For $t > 1$, conditioning on x_0 yields a Gaussian posterior with

$$\begin{aligned} \tilde{\beta}_t &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t, \\ \tilde{\mu}_t &= \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t. \end{aligned} \tag{35}$$

With fixed reverse variance $\sigma_t^2 > 0$, the Gaussian KL is $\mathbb{E}[\|\tilde{\mu}_t - \mu_\theta\|^2 / (2\sigma_t^2)] + C$. Substituting $x_0 = (x_t - \sqrt{1 - \bar{\alpha}_t}\varepsilon)/\sqrt{\bar{\alpha}_t}$ gives

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right). \tag{36}$$

Replacing ε by ε_θ defines the model mean in Eq. (10); subtracting the two means gives

$$w_t = \frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)}. \tag{37}$$

These are DDPM's Eqs. 8–12. The simplified objective omits this coefficient for intermediate steps; its $t = 1$ term is related to the endpoint decoder using the approximation explained in DDPM Section 3.4. Consequently L_{simple} is a reweighted surrogate, not numerically the original variational bound. The CIFAR-10 results in Section 5.2 empirically motivate that change.

C.7. Why the score determines the reverse mean

Apply the marginal-score identity to a single Gaussian forward step:

$$\nabla \log q_t(x_t) = -\frac{x_t - \sqrt{\alpha_t}\mathbb{E}[x_{t-1} \mid x_t, t]}{\beta_t}. \tag{38}$$

Therefore its true reverse conditional mean is

$$\mathbb{E}[x_{t-1} \mid x_t, t] = \frac{x_t + \beta_t \nabla \log q_t(x_t)}{\sqrt{\alpha_t}}. \tag{39}$$

Using the estimated DDPM score $-\varepsilon_\theta(x_t, t)/\sqrt{1-\bar{\alpha}_t}$ gives the reverse mean in Eq. (10). The conditional-mean identity is exact under the stated assumptions, but it does not make the full reverse conditional Gaussian. Its higher moments and possible multimodality are not specified by the mean alone. DDPM uses a Gaussian approximation motivated by small forward steps and learns an approximate score.

C.8. Why noise prediction is useful without knowing the true noise

For a fixed x_t and t, the vector minimizing expected squared prediction error is $\mathbb{E}[\varepsilon \mid x_t, t]$ (and, in a conditional model, $\mathbb{E}[\varepsilon \mid x_t, t, y]$). Thus a trained predictor estimates a statistical direction shared by plausible clean examples, not an oracle that recovers every random draw. Its corresponding clean-state estimate is

$$\hat{x}_0(x_t, t) = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\varepsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \tag{40}$$

This follows by rearranging Eq. (2); prediction errors remain and can be magnified at small $\bar{\alpha}_t$. The reverse sampler uses predictions over many levels rather than treating one estimate as guaranteed recovery. The same argument applies in latent space, followed by the learned decoder.

C.9. Two training stages and their different roles

Stage one updates encoder ϕ and decoder ψ to preserve useful image information in a regularized representation. Stage two fixes them and trains denoiser θ with Eq. (19). The original LDM conditional formulation jointly optimizes the condition encoder and denoiser [10, Sec. 3.3]; particular later implementations may instead keep the text encoder fixed.

The distinction matters: improving the denoiser cannot guarantee exact recovery of details that the first stage systematically discards. Conversely, a high-quality autoencoder does not itself provide the conditional latent distribution needed for generation. The VAE prior regularizer and the learned diffusion prior play different roles.

D. Supporting Autoencoder and Conditioning Details

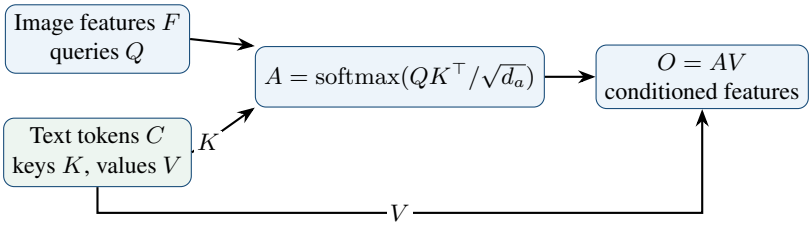


Figure 4. Original single-head cross-attention schematic, based on [10, 17]. Multi-head attention learns several such projections and combines their outputs.

D.1. Textbook VAE objective and the LDM distinction

A textbook negative evidence lower bound for a VAE is [8]

$$\begin{aligned}\mathcal{J}_{\mathrm{VAE}} = \mathbb{E}_{q_\phi(\tilde{z}|x_0)}[-\log p_\psi(x_0 \mid \tilde{z})] \\ + \mathrm{D}_{\mathrm{KL}}(q_\phi(\tilde{z} \mid x_0) \| \mathcal{N}(0, \mathbf{I}))\,.\end{aligned} \tag{41}$$

The decoder likelihood $p_\psi(x_0 \mid \tilde{z})$ rewards reconstruction; the nonnegative KL term regularizes the approximate posterior toward the prior. Natural logarithms give nats. This explains the VAE interpretation, but LDM's perceptual/adversarial first-stage objective with weak KL regularization is not identical to this pixel-likelihood objective [10, Sec. 3.1].

D.2. Cross-attention: how language reaches the denoiser

Use a *row-vector* convention. At one U-Net layer, flatten its feature map into $F \in \mathbb{R}^{n\times d_u}$, where n is the number of spatial positions and d_u is its channel width. Let $C = \tau_\eta(y) \in \mathbb{R}^{m\times d_y}$ contain m token embeddings of width d_y. One cross-attention head computes [10, 17]

$$Q = FW_Q, \quad K = CW_K, \quad V = CW_V, \tag{42}$$

$$A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_a}}\right), \qquad O = AV. \tag{43}$$

The learned projections have shapes $W_Q \in \mathbb{R}^{d_u\times d_a}$, $W_K \in \mathbb{R}^{d_y\times d_a}$, and $W_V \in \mathbb{R}^{d_y\times d_v}$. Thus $Q \in \mathbb{R}^{n\times d_a}$ contains *queries from image features*, $K \in \mathbb{R}^{m\times d_a}$ contains *keys from text*, and $V \in \mathbb{R}^{m\times d_v}$ contains text values. d_a, d_v are positive integer head widths. A superscript $\top$ denotes transposition. Softmax acts across the m tokens for each spatial row, making $A \in \mathbb{R}^{n\times m}$ nonnegative with row sums of one. $O \in \mathbb{R}^{n\times d_v}$ supplies text-dependent features that are projected back into the U-Net's residual stream.

A numerical attention example

Take one spatial query $Q = (1, 0)$, two keys $K_1 = (\sqrt{2}, 0)$, $K_2 = (0, 0)$, and $d_a = d_v = 2$. The scaled logits are $(1, 0)$, so $A = (e/(e + 1), 1/(e + 1)) \approx (0.7311, 0.2689)$, where e is Euler's number. For $V_1 = (1, 0)$ and $V_2 = (0, 1)$, the output is $O \approx (0.7311, 0.2689)$. This original toy example shows weighted retrieval, not an observed attention map from an actual model.

D.3. Prompt control and classifier-free guidance

Cross-attention supplies the condition; *classifier-free guidance* (CFG) adjusts how strongly a conditional prediction influences sampling [5]. Train the same network with the condition replaced by a null condition $\varnothing$ with probability $p_{\mathrm{drop}} \in (0, 1)$. At inference define

$$\begin{aligned}\varepsilon_u &= \varepsilon_\theta(z_t, t, \tau_\eta(\varnothing)), \\ \varepsilon_c &= \varepsilon_\theta(z_t, t, \tau_\eta(y)), \\ \varepsilon_{\mathrm{cfg}} &= \varepsilon_u + s(\varepsilon_c - \varepsilon_u).\end{aligned} \tag{44}$$

$s \geq 0$ is a dimensionless guidance scale. Here $s = 0$ uses the unconditional prediction, $s = 1$ gives ordinary conditional sampling, and $s > 1$ extrapolates in the conditional direction. The CFG paper writes the equivalent expression $(1 + w)\varepsilon_c - w\varepsilon_u$; its parameter is $w = s - 1$. Stating the convention prevents confusing two scales with different numerical baselines.

For an original scalar illustration, $\varepsilon_u = 0.2$, $\varepsilon_c = 0.1$ and $s = 3$ give $\varepsilon_{\mathrm{cfg}} = -0.1$. Guidance is therefore an extrapolation, not a convex average. Substituting it into the reverse update changes the generation trajectory. Greater guidance can improve conditional fidelity while reducing diversity; it does not ensure monotonically better quality [5]. Standard CFG requires two denoiser evaluations per step, often batched together, adding compute and potentially peak memory.

For a proposed prompt-control experiment, including fixed seeds and separate attribute/relation changes, see Appendix B.

E. Qualitative Comparison of Generative Models

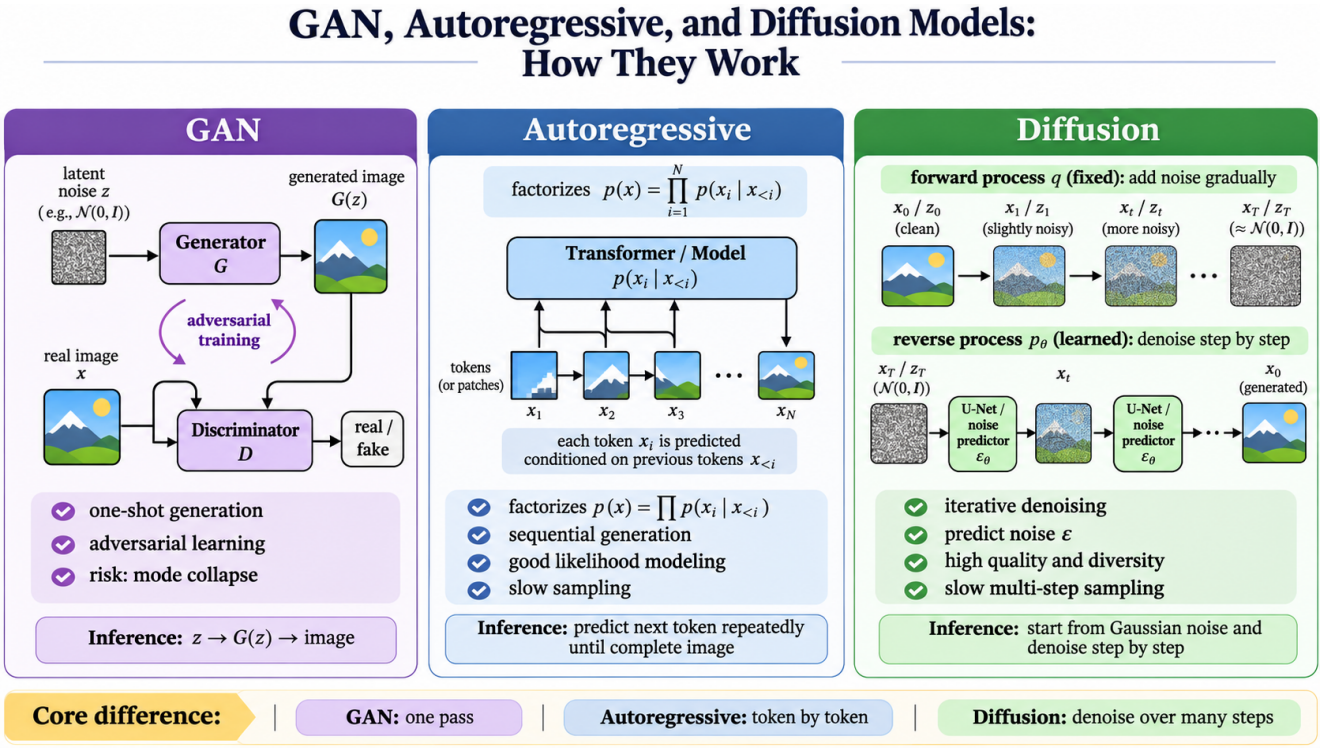


Figure 5. Overview of GAN, autoregressive, and diffusion models. Related methods are described in [3, 6, 16]. The labels describe typical behavior rather than universal performance guarantees.

F. Pseudocode for Forward and Backward Flows

These algorithms make the Gaussian processes in Sections 3.1–3.2 explicit, using the 2015 reverse-parameter formulation [12]. All states lie in $\mathbb{R}^D$; each Gaussian noise draw is independent. The two procedures describe corruption and generation; the DDPM variant is distinguished below.

F.1. Forward noising

Algorithm 1 Forward Gaussian diffusion

Input: Clean state x_0; schedule $\beta_1, \dots, \beta_T$
Output: Forward trajectory $(x_0, x_1, \dots, x_T)$

for $t = 1, \dots, T$ **do**
  Draw $\varepsilon^{(t)} \sim \mathcal{N}(0, \mathbf{I})$
  $\alpha_t \leftarrow 1 - \beta_t$
  $x_t \leftarrow \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon^{(t)}$
end for
return $(x_0, x_1, \dots, x_T)$

What this computes. Algorithm 1 applies the known Gaussian transition in Eq. (1). The schedule satisfies $0 < \beta_t < 1$; the original Gaussian framework may learn its rates, whereas the reported DDPM implementation fixes them. The forward transition itself does not evaluate a denoising network. A sufficiently long schedule makes the terminal distribution approximately standard Gaussian, not necessarily exactly Gaussian at finite T.

Direct access to one timestep. When only one noisy training input is required, the full loop is unnecessary. Compute $\bar{\alpha}_t = \prod_{j=1}^{t}(1 - \beta_j)$ and draw

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathbf{I}).$$

This samples the same marginal $q(x_t \mid x_0)$ as Eq. (2); it does not recover the particular path generated by Algorithm 1.

Separate runs and separate noise. Backward generation starts from a fresh prior draw, independently of the clean image and trajectory used in Algorithm 1. The reverse noise u_t is not the forward innovation $\varepsilon^{(t)}$. Composing the learned reverse transitions produces a model sample; it does not guarantee recovery of the original x_0.

F.2. Backward sampling: the 2015 Gaussian model

Algorithm 2 Backward Gaussian sampling

Input: T; trained f_μ, f_Σ; first-step variance β_1
Output: A generated state x_0

Draw $x_T \sim \mathcal{N}(0, \mathbf{I})$
for $t = T, T-1, \dots, 2$ **do**
  $\mu_t \leftarrow f_\mu(x_t, t)$
  $\Sigma_t \leftarrow f_\Sigma(x_t, t)$
  Draw $u_t \sim \mathcal{N}(0, \mathbf{I})$
  $x_{t-1} \leftarrow \mu_t + \Sigma_t^{1/2} u_t$
end for
Draw $u_1 \sim \mathcal{N}(0, \mathbf{I})$
$x_0 \leftarrow \sqrt{1 - \beta_1}\, x_1 + \sqrt{\beta_1}\, u_1$
return x_0

Learned intermediate steps. The network receives the current noisy state and timestep, and predicts a reverse mean and positive covariance. For the original image model, Σ_t is diagonal and its square root is computed elementwise. The random draw implements $p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_t, \Sigma_t)$, as in Eq. (4).

The 2015 endpoint. The last update uses the prescribed stationary Gaussian kernel [12, App. B.2], rather than another learned transition. Its formula and role in the original variational bound are given in Appendix C.5.

How DDPM changes the backward sampler. DDPM [6, Algorithm 2] computes μ_θ from ε_θ using Eq. (10) and uses a fixed $\sigma_t^2 \mathbf{I}$ for the intermediate reverse covariance. The two choices, $\sigma_t^2 = \beta_t$ and $\sigma_t^2 = \tilde{\beta}_t$, are specified in Eq. (14). Its update is $x_{t-1} = \mu_\theta(x_t, t) + \sigma_t u_t$ for $t = T, \dots, 1$, drawing $u_t \sim \mathcal{N}(0, \mathbf{I})$ when $t > 1$ and setting $u_1 = 0$ for the final update. Thus the 2015 endpoint in Algorithm 2 must also be replaced when implementing DDPM. Forward noising remains Algorithm 1; Sections 3.4 and 3.6 explain the parameterization and sampling changes.